Natasha E. Batalha

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Studies planetary atmospheres at the nexus of observation and theory, with planets within the Solar System and beyond. Leverages and develops open science theoretical models (climate, spectroscopic, cloud) to determine atmospheric properties from spectroscopic observations of exoplanets and Brown Dwarfs.

Appointments

- **NASA Ames Research Center** — **Moffett Field, CA**
  Research Scientist — *Oct 2019 – present*
- **University of California Santa Cruz** — **Santa Cruz, CA**
  UC President's Postdoctoral Fellow — *Sep 2018 – Sep 2019*
- **Space Telescope Science Institute** — **Baltimore, MD**
  Postdoc Science Mission Office — *Aug 2017 - Aug 2018*

Education

- **The Pennsylvania State University** — **State College, PA**
  Dual PhD, Astronomy/Astrophysics & Astrobiology — *2017*
  Dissertation: A Synergistic Approach to Interpreting Planetary Atmospheres
- **Cornell University** — **Ithaca, NY**
  B.A., Physics — *2013*

Awards, Fellowships

- **2025**: Presidential Early Career Award for Scientists and Engineers: Given by President Biden
- **2025**: NASA Group Achievement Award for the Habitable Worlds Observatory Collaborative Team
- **2025**: NASA Group Achievement Award for the Chabot Space Internship Mentors
- **2024**: NASA Honor Award: Early Career Achievement Medal
- **2023**: NASA Ames Peer-Nominated Tim Lee Outstanding Early Career Space Scientist Lecture Award
- **2022**: NASA Group Achievement Award for the ExoExplorers Program Team
- **2021**: NASA Ames Research Center Early Career Award
- **2020**: Evans Visiting Lectureship in Exoplanet Science, UC Irvine
- **2018**: University of California President's Postdoctoral Fellowship
- **2017**: Ford Foundation Fellow, Honorable Mention
- **2017**: Alfred P. Sloan Foundation Minority Graudate Scholarship
- **2016**: Kavli Student Fellow
- **2015**: National Astrobiology Early Career Collaboration Award
- **2015**: Stephen B. Brumback Graduate Fellowship in Astrophysics

- **2014**: National Science Foundation Graduate Research Fellowship
- **2013**: STEM Scholar Graduate Fellow

Open Science Projects | `GitHub:` ☆170 ⑂128 ⭳7,477 | `Zenodo:` ⭳16,924

- `PICASO`: `https://natashabatalha.github.io/picaso`
  Enables computation of reflected light, thermal, and transmission spectroscopy for exoplanets and Brown Dwarfs.
- `PandExo`: `https://natashabatalha.github.io/PandExo`
  Enables simulations of JWST and HST observations.
- `Virga`: `https://natashabatalha.github.io/Virga`
  Enables theoretical modeling of exoplanet and Brown Dwarf clouds.
- `MAESTRO`: `https://science.data.nasa.gov/opacities/`
  Enables broad access to critical opacity data needed for analysis of substellar atmospheres.

Awarded Grants & Observing Time

- **2025**: **PI** | NASA Internal Science Funding Model
  Line Lists and Opacities for Enabling Astrophysics Discoveries
- **2025**: **Co-I** | JWST-GO-11962
  Twinning: An Atmospheric Study of Twin Sub-Neptunes To Investigate Carbon Chemistry, Aerosol Formation, and Photochemistry. PI: Scarsdale, N
- **2024**: **Co-I** | JWST-GO-09256
  Using stellar contamination proxy TRAPPIST-1 b to search for an atmosphere on TRAPPIST-1 e. PI: Allen, N
- **2024**: **Co-I** | JWST-GO-09033
  By the Ashes of Stars: A Chemical Census of a White Dwarf Planet. PI: MacDonald,R
- **2024**: **Co-I** | JWST-AR-08657
  How does Rotation Impact Y Dwarf Atmospheres?. PI: Lacy, B
- **2024**: **Co-I** | JWST-GO-08309
  It's (poly)Morphin? Time! Solving the quartz quandary of WASP-17b. PI: Mullens, E
- **2024**: **PI** | JWST-GO-08004
  Cliff Hangers: Testing for Atmosphere-Mantle Interactions in Radius Cliff Planets.
- **2024**: **Co-I** | JWST-GO-07849
  Meeting their Match: Measuring Cloudy Hot Jupiter Counterparts to Understand the Silicate Cloud Regime with JWST/MIRI. PI: Moran, S
- **2024**: **Co-I** | JWST-AR-07358
  The Other Extreme: Enabling Characterization of Metal-poor Brown Dwarfs and testing our Understanding of Jupiter and Saturn. PI: Mukherjee, S
- **2024**: **Co-I** | JWST-GO-07073
  Charting the Cosmic Shoreline. PI: Lustig-Yaeger
- **2024**: **Co-I** | JWST-GO-06456
  Using stellar contamination proxy TRAPPIST-1 b to search for an atmosphere on TRAPPIST-1 e. PI: Allen, N
- **2024**: **Co-I** | HST-GO-17801 | 29 orbits
  Characterizing the Detected Atmospheric Escape of a Pair of Small Sub-Neptunes. Co-PI: Loyd, Parke
- **2024**: **Co-I** | 24-SOSS24-0023

The First Set of Documentation and Tutorials for Any Photochemical Exoplanet Code. PI: Wogan, Nick

- **2024**: **Collaborator** | 24-HW24

  Exploring the Influence of Hazes on Climate and Spectra of Small Planets around FGK Stars. PI: Kopparapu, Ravi
- **2023**: **PI** | HST-AR-17418 | 20 orbits

  Upon Reflection: Unveiling a Cloud Transition with a High-Precision Reflected Light Spectrum. Co-PI: Wakeford, Hannah
- **2023**: **Co-PI** | HST-AR-17558

  Sulfur Vision! Refining our view of sulfur-containing UV opacities for exoplanet atmospheres. PI: Hargreaves, Rob
- **2023**: **Co-I** | 23-XRP23-0107

  From Super-Earths to Sub-Neptunes: Laboratory Measurements of High-Temperature Absorption Cross Sections and Pressure-broadening coefficients of key exoplanetary molecules with High Resolution Mid-IR Spectroscopy. PI: Bejaoui, Salma
- **2023**: **Collaborator** | 23-XRP23-0107

  Taking Water Worlds Seriously: Exploring the Phase Space of Water World Atmospheres, Structure, and Evolution. PI: Fortney, Jonathan
- **2023**: **Co-I** | 23-XRP23-0081

  The Bridge to Jupiter: Modeling Cool Gas Giants in the JWST Era. PI: Morley, Caroline
- **2023**: **Co-I** | JWST-AR-3245

  Up to the Task? A New Generation of Atmospheric and Interior Models of Brown Dwarfs for the JWST Era. PI: Mukherjee, S.
- **2023**: **Co-I** | JWST-AR-3207

  Lifting the Veil: An Open Source Haze Model for Exoplanet Atmospheric Characterization. PI: Gao, P.
- **2023**: **Co-I** | JWST-AR-3201

  The Utility of Self-Consistent Models and Photochemistry in Understanding Transiting Planet Atmospheres. PI: Fortney, J.
- **2023**: **Co-I** | Chandra-DDT

  A Chandra & JWST View of M-Dwarf Rocky Planet Atmospheres. PI: Howard, W.
- **2023**: **Co-I** | 22-XRP22-2-0083

  Estimating pi with PIE: Constraining the Population Proportion of M-Dwarf Planetary Atmospheres with Planetary Infrared Excess. PI: Stevenson, K.
- **2023**: **Co-I** | 22-TCAN22-0033

  Accelerating Substellar Atmosphere Spectral Inference with Machine Learning Technologies. PI: Gully-Santiago, Michael
- **2022**: **Co-I** | 22-ICAR22-0001

  The Virtual Planetary Laboratory: Advancing the Search for Life on Exoplanets.. PI: Meadows, Victoria
- **2022**: **Collaborator** | 22-PDAR22-0025

  Enhancing Capabilities of the HITRAN and HITEMP Molecular Spectroscopic Databases for Studies of Planetary Atmospheres. PI: Gordon, Iouli
- **2022**: **Co-I** | HST-GO-17183| 122 orbits

  Hubble Ultraviolet-optical Survey of Transiting Legacy Exoplanets (HUSTLE) treasury program. PI: Wakeford, H.
- **2021**: **PI** | 21-XRP21-0182

  Towards High Metallicity: Integrated Composition-Dependent Molecular Opacities for Modeling Super-Earth to Neptunian Atmospheres. Science PI: Gharib-Nezhad, Ehsan
- **2021**: **PI** | JWST-GO-2512 | 142 hours

Seeing the Forest and the Trees: Unveiling Small Planet Atmospheres with a Population-Level Framework. Co-PI: Teske, Johanna

- **2021**: Co-I | JWST-GO-2358 | 13.1 hours
  Under the Light of a Dead Star: Revealing the Atmospheric Composition of a White Dwarf Planet. PI: MacDonald, Ryan
- **2021**: Co-I | JWST-GO-2358 | 75.6 hours
  Tell Me How I'm Supposed To Breathe With No Air: Measuring the Prevalence and Diversity of M-Dwarf Planet Atmospheres. PI: Stevenson, Kevin
- **2021**: Co-I | JWST-GO-2667 | 9.2 hours
  Good vibrations: Directly measuring Exoplanet aerosol compositions with MIRI spectroscopy. PI: Wakeford, Hannah
- **2021**: Co-I | JWST-AR-1977
  Glows in the Dark: New Models for the Atmospheric Structure and Evolution of High Metallicity and. PI: Marley, Mark
- **2021**: Subject Level Member | JWST-GTO-1353 | 74.9 hours
  Transit and Eclipse Spectroscopy of a Hot Jupiter. PI: Lewis, N.
- **2020**: Subject Level Member | JWST-GTO-1312 | 34.1 hours
  Transiting and Eclipse Spectroscopy of a Warm Neptune. PI: Lewis, N.
- **2020**: Subject Level Member | JWST-GTO-1331 | 22 hours
  Transit Spectroscopy of TRAPPIST-1e. PI: Lewis, N.
- **2020**: Co-I | 19-ICAR19-0009
  The M-dwarf Opportunity: Characterizing Nearby M-dwarf Habitable Zone Planets. PI: Stevenson, Kevin.
- **2020**: Co-I | 19-ICAR19-0043
  Follow the Volatiles: Tracing chemical species relevant to habitability from proto-planetary disks to exoplanet atmospheres. PI: Batalha, N.M.
- **2020**: Co-I | HST-GO-16180
  Constructing the First Spectroscopic, Multi-Dimensional Map of a Hot Jupiter. PI: Kataria, T
- **2020**: Co-I | Gemini 2020-LP
  A high-resolution survey of molecular abundances in transiting exoplanet atmospheres. PI: Mansfield, M.
- **2019**: Co-I | HST-GO-15836
  A deep look into the atmosphere of an exoplanet around a pre-main sequence star. PI: Newton, E.
- **2019**: Collaborator | Planetary Data Archiving, Restoration, and Tools
  Enhancing capabilities of the HITRAN and HITEMP molecular spectroscopic databases for planetary research. PI:Gordon, I.
- **2019**: **Science PI** | NASA Unsolicited Proposal
  Community Tool for Computing, Manipulating and Visualizing Molecular and Atomic Opacities. PI:Lewis, N.K.
- **2017**: Co-I | JWST-ERS-1366
  The Transiting Exoplanet Community Early Release Science Program. PI: Batalha, N.M.
- **2017**: Co-I | HST-GO-14918
  Definitive Measurement of WASP-17b's Water Abundance in Preparation of JWST. PI: Wakeford, H.R.

Professional Service

Referee: AAS, MNRAS
Panelist:TESS, HST, ROSES

Committees: 2024-p: Science Advisory Committee for STScI's DDT Rocky Worlds Program
2020-p: Planetary Data System User Committee
2023-'24: START Habitable Worlds Observatory
2023-'24: Strategic Exoplanet Initiative with HST-JWST Working Group
2019-'23: ExoPAG Executive Committee

Organizer: 2020-'23: ExoExplorers Program
2015: AbGradCon
2014: Emerging Researchers in Exoplanet Symposium

Lead: 2021-p: Bay Area Exoplanet Meeting
2022-'23: Open Source Science Community Building: Module Lead for Open Results

Broader Impacts

- **2024-2025**: Career Mentor, Latino Education Advancement Foundation*

  LEAF's mission is to improve outcomes for Latinx students and families in East San José by providing support services for college enrollment and persistence
- **2021-p**: Subject Matter Expert, Chabot Space Science Center *

  Regular speaker on topics related to NASA's search for life beyond year, women in STEM, and diversity, equity and inclusion within STEM
- **2021-p**: Subject Matter Expert, NASA Community College Network

  An initiative to bring NASA Subject Matter Experts (SMEs), research findings, and science resources into the nation's community college system
- **2018-2020**: Advisor/Instructor, Evergreen Valley Community College - Citizen Science Initiative*

  501(c)3 with the goal of increasing BIPOC students in STEM.
- **2017**: Instructor, Project Favela

  * *501(c)3 with the goal of providing education to students in Rocinha, one of Brazil's largest favelas.*
- **2014-2017**: Instructor, Centre County Prison Society Education Program

  * *501(c)3 with the goal of providing education within the prison system.*
- **2015-2017**: Director of Programs, Learn to Be Foundation

  * *501(c)3 with the goal of providing underserved K-12 students with free 1-on-1 online tutoring*

* denotes personal time

Research in Media

- **NASA News | Feb. 2024** : NASA Astronomer Sees Power in Community, Works to Build More. See link at nasa.gov
- **IMAX Documentary | Oct. 2023** : Deep Sky `https://www.imdb.com/title/tt28370567/`
- **NASA News | Oct. 2023** : Webb Detects Tiny Quartz Crystals in the Clouds of a Hot Gas Giant link at www.nasa.gov

  . Reported in Daily Express US, The Conversation, and more
- **NASA News | Jan. 2023** : NASA's Webb Confirms Its First Exoplanet link at www.jpl.nasa.gov

  . Reported in CNN, NBC, and more
- **NASA News | Aug. 2023** : NASA's Webb Detects Carbon Dioxide in Exoplanet Atmosphere link at www.jpl.nasa.gov

  . Reported in CNN, NY Times, and more
- **Nature Career | Feb. 2023** : Mother–daughter duo work together to find new worlds `https://www.nature.com/articles/d41586-023-00580-6`
- **Clear+Vivid with Alan Alda | Mar. 2022** : Podcast Interview, "Natalie and Natasha Batalha: Looking for Life on Alien Worlds" `https://www.youtube.com/watch?v=EKfHnNWo9hc`

- **Vice News** | **Dec. 2021** | **15k+ views**: The World's Largest Telescope is Leaving Earth https://www.youtube.com/watch?v=aqI_HXo_0gs
- **Quanta Magazine Mini-Documentary** | **Dec. 2021** | **2M+ views**: "The Webb Space Telescope Will Rewrite Cosmic History" https://www.youtube.com/watch?v=shPwW11MEHg
- **Quanta Magazine Feature** | **Dec. 2021**: "The Webb Space Telescope Will Rewrite Cosmic History. If It Works." link at www.quantamagazine.org
- **Emmy Winning CNN Documentary** | **2021**: "The Hunt for Planet B" Directed by Nathaniel Kan https://www.imdb.com/title/tt13848014/
- **UCSC Press** | **2021**: "Meet the NASA scientist opening up exoplanet research" link at ucscsciencenotes.com
- **NBC News** | **Dec. 2021** | **15k+ views**: "Silicon Valley Scientists Tout Historic Launch of James Webb Space Telescope" https://www.youtube.com/watch?v=EKfHnNWo9hc
- **AirTalk with KPCC NPR** | **Dec. 2021** : "NASA Prepares to Launch Its Most Complicated Telescope to Date: The James Webb" link at www.kpcc.org
- **National Geographic** | **Dec. 2021** : "The James Webb Space Telescope Will Transform our Understanding of Alien Worlds" link at www.nationalgeographic.com
- **Canadian Broadcast Corporation, The Current with Matt Galloway** | **Dec. 2021**: "The James Webb Space Telescope Gets Ready for Lift-Off" link at www.cbc.ca
- **NPR KQED News** | **Apr. 2021** | **1k+ views**: "Search for Exoplanets" https://www.youtube.com/watch?v=HD_02wsYlyU&t=921s

Publications | h-index:45 | i10-index: 100 | Citations:8554

Link to all publications

Selected First Author or Major Contributor

- Wogan, N. F., Batalha, N. E., Krissansen-Totton, J., et al. 2026, Toward Inferring the Surface Fluxes of Biosignature Gases on Rocky Exoplanets from Telescope Spectra, Astrophys. J., doi.org/10.3847/1538-4357/ae6646
- Lew, B. W. P., Roellig, T., Batalha, N. E., et al. 2026, JWST Spectral Retrieval of Cold Directly Imaged Planet WD 0806 b and the First Measurement of Altitude-dependent Kzz in Exoplanet Atmospheres, Astron. J., doi.org/10.3847/1538-3881/ae4747
- Mang, J., Batalha, N. E., Morley, C. V., et al. 2026, PICASO 4.0: Clouds and Photochemistry in Climate Models of Brown Dwarfs and Exoplanets, Astrophys. J., doi.org/10.3847/1538-4357/ae47ff
- Gordon, T. A., Batalha, N. M., Batalha, N. E., et al. 2026, JWST COMPASS: Insights into the Systematic Noise Properties of NIRSpec/G395H from a Uniform Reanalysis of Seven Transmission Spectra, Astron. J., doi.org/10.3847/1538-3881/ae3de9
- Batalha, N. E., Rooney, C. M., Visscher, C., et al. 2026, Condensation Clouds in Substellar Atmospheres with Virga, Astron. J., doi.org/10.3847/1538-3881/ae29e5
- Vaughan, S. R., Birkby, J. L., Batalha, N. E., et al. 2026, No TiO detected in the hot-Neptune-desert planet LTT-9779 b in reflected light at high spectral resolution, Astron. Astrophys., doi.org/10.1051/0004-6361/202557240
- Wogan, N. F., Batalha, N. E., Zahnle, K., et al. 2025, The Open-source Photochem Code: A General Chemical and Climate Model for Interpreting (Exo)Planet Observations, Planet. Sci. J., doi.org/10.3847/PSJ/ae0e1c
- Moran, S. E., Lodge, M. G., Batalha, N. E., et al. 2025, Fractal Aggregate Aerosols in the Virga Cloud Code. I. Model Description and Application to a Benchmark Cloudy Exoplanet, Astrophys. J., doi.org/10.3847/1538-4357/ae0583
- Dressier, A., Batalha, N. E., Wogan, N., et al. 2025, JWST-TST DREAMS: Sulfur Dioxide in

the Atmosphere of the Neptune-mass Planet HAT-P-26 b from NIRSpec G395H Transmission Spectroscopy, Astron. J., doi.org/10.3847/1538-3881/ae0929

- Teske, J., Batalha, N. E., Wallack, N. L., et al. 2025, JWST COMPASS: NIRSpec/G395H Transmission Observations of TOI-776 c, a 2 R M Dwarf Planet, Astron. J., doi.org/10.3847/1538-3881/adb975
- Gharib-Nezhad, E. (Sam) ., Valizadegan, H., Batalha, N. E., Martinho, M. J. S., Lew, B. W. P. 2025, TelescopeML. II. Convolutional Neural Networks for Predicting Brown Dwarf Atmospheric Parameters, Astrophys. J., doi.org/10.3847/1538-4357/ada1d2
- Hamill, C. D., Johnson, A. V., Batalha, N., et al. 2024, Reflected-light Phase Curves with PICASO: A Kepler-7b Case Study, Astrophys. J., doi.org/10.3847/1538-4357/ad7de6
- Inglis, J., Batalha, N. E., Lewis, N. K., et al. 2024, Quartz Clouds in the Dayside Atmosphere of the Quintessential Hot Jupiter HD 189733 b, Astrophys. J. Lett., doi.org/10.3847/2041-8213/ad725e
- Wallack, N. L., Batalha, N. E., Alderson, L., et al. 2024, JWST COMPASS: A NIRSpec/G395H Transmission Spectrum of the Sub-Neptune TOI-836c, Astron. J., doi.org/10.3847/1538-3881/ad3917
- Gharib-Nezhad, E. (Sam) ., Batalha, N., Valizadegan, H., et al. 2024, TelescopeML - I. An End-to-End Python Package for Interpreting Telescope Datasets through Training Machine Learning Models, Generating Statistical Reports, and Visualizing Results, The Journal of Open Source Software, doi.org/10.21105/joss.06346
- Lew, B. W. P., Roellig, T., Batalha, N. E., et al. 2024, High-precision Atmospheric Characterization of a Y Dwarf with JWST NIRSpec G395H Spectroscopy: Isotopologue, C/O Ratio, Metallicity, and the Abundances of Six Molecular Species, Astron. J., doi.org/10.3847/1538-3881/ad3425
- Alderson, L., Batalha, N. E., Wakeford, H. R., et al. 2024, JWST COMPASS: NIRSpec/G395H Transmission Observations of the Super-Earth TOI-836b, Astron. J., doi.org/10.3847/1538-3881/ad32c9
- Wogan, N. F., Batalha, N. E., Zahnle, K. J., et al. 2024, JWST Observations of K2-18b Can Be Explained by a Gas-rich Mini-Neptune with No Habitable Surface, Astrophys. J. Lett., doi.org/10.3847/2041-8213/ad2616
- Gharib-Nezhad, E. (Sam) ., Batalha, N. E., Chubb, K., et al. 2024, The impact of spectral line wing cut-off: recommended standard method with application to MAESTRO opacity data base, RAS Techniques and Instruments, doi.org/10.1093/rasti/rzad058
- Rooney, C. M., Batalha, N. E., Marley, M. S. 2024, Spherical Harmonics for the 1D Radiative Transfer Equation. II. Thermal Emission, Astrophys. J., doi.org/10.3847/1538-4357/ad05c5
- Rooney, C. M., Batalha, N. E., Marley, M. S. 2023, Spherical Harmonics for the 1D Radiative Transfer Equation. I. Reflected Light, Astrophys. J., doi.org/10.3847/1538-4357/acca79
- Mukherjee, S., Batalha, N. E., Fortney, J. J., Marley, M. S. 2023, PICASO 3.0: A One-dimensional Climate Model for Giant Planets and Brown Dwarfs, Astrophys. J., doi.org/10.3847/1538-4357/ac9f48
- Batalha, N. E., Wolfgang, A., Teske, J., et al. 2023, Importance of Sample Selection in Exoplanet-atmosphere Population Studies, Astron. J., doi.org/10.3847/1538-3881/ac9f45
- Mukherjee, S., Fortune, J. J., Batalha, N. E., et al. 2022, Probing the Extent of Vertical Mixing in Brown Dwarf Atmospheres with Disequilibrium Chemistry, Astrophys. J., doi.org/10.3847/1538-4357/ac8dfb
- Robbins-Blanch, N., Kataria, T., Batalha, N. E., Adams, D. J. 2022, Cloudy and Cloud-free Thermal Phase Curves with PICASO: Applications to WASP-43b, Astrophys. J., doi.org/10.3847/1538-4357/ac658c
- Adams, D. J., Kataria, T., Batalha, N. E., Gao, P., Knutson, H. A. 2022, Spatially Resolved Modeling of Optical Albedos for a Sample of Six Hot Jupiters, Astrophys. J., doi.org/10.3847/1538-4357/ac3d32
- Rooney, C. M., Batalha, N. E., Gao, P., Marley, M. S. 2022, A New Sedimentation Model for Greater

Cloud Diversity in Giant Exoplanets and Brown Dwarfs, Astrophys. J., doi.org/10.3847/1538-4357/ac307a
- Gharib-Nezhad, E., Marley, M. S., Batalha, N. E., et al. 2021, Following the Lithium: Tracing Li-bearing Molecules across Age, Mass, and Gravity in Brown Dwarfs, Astrophys. J., doi.org/10.3847/1538-4357/ac0a7d
- Mukherjee, S., Batalha, N. E., Marley, M. S. 2021, Cloud Parameterizations and their Effect on Retrievals of Exoplanet Reflection Spectroscopy, Astrophys. J., doi.org/10.3847/1538-4357/abe53b
- Batalha, N. E., Lewis, T., Fortney, J. J., et al. 2019, The Precision of Mass Measurements Required for Robust Atmospheric Characterization of Transiting Exoplanets, Astrophys. J. Lett., doi.org/10.3847/2041-8213/ab4909
- Batalha, N. E., Marley, M. S., Lewis, N. K., Fortney, J. J. 2019, Exoplanet Reflected-light Spectroscopy with PICASO, Astrophys. J., doi.org/10.3847/1538-4357/ab1b51
- Moran, S. E., Hörst, S. M., Batalha, N. E., Lewis, N. K., Wakeford, H. R. 2018, Limits on Clouds and Hazes for the TRAPPIST-1 Planets, Astron. J., doi.org/10.3847/1538-3881/aae83a
- Batalha, N. E., Smith, A. J. R. W., Lewis, N. K., et al. 2018, Color Classification of Extrasolar Giant Planets: Prospects and Cautions, Astron. J., doi.org/10.3847/1538-3881/aad59d
- Batalha, N. E., Lewis, N. K., Line, M. R., Valenti, J., Stevenson, K. 2018, Strategies for Constraining the Atmospheres of Temperate Terrestrial Planets with JWST, Astrophys. J. Lett., doi.org/10.3847/2041-8213/aab896
- Batalha, N. E., Kopparapu, R. K., Haqq-Misra, J., Kasting, J. F. 2018, Reply to Shaw, Earth and Planetary Science Letters, doi.org/10.1016/j.epsl.2017.12.018
- Batalha, N. E., Mandell, A., Pontoppidan, K., et al. 2017, PandExo: A Community Tool for Transiting Exoplanet Science with JWST HST, Publ. Astron. Soc. Pac., doi.org/10.1088/1538-3873/aa65b0
- Batalha, N. E., Line, M. R. 2017, Information Content Analysis for Selection of Optimal JWST Observing Modes for Transiting Exoplanet Atmospheres, Astron. J., doi.org/10.3847/1538-3881/aa5faa
- Batalha, N. E., Kempton, E. M.-R., Mbarek, R. 2017, Challenges to Constraining Exoplanet Masses via Transmission Spectroscopy, Astrophys. J. Lett., doi.org/10.3847/2041-8213/aa5c7d
- Batalha, N. E. 2017, A Synergistic Approach to Interpreting Planetary Atmospheres, Ph.D. Thesis, doi.org/
- Batalha, N. E., Kopparapu, R. K., Haqq-Misra, J., Kasting, J. F. 2016, Climate cycling on early Mars caused by the carbonate-silicate cycle, Earth and Planetary Science Letters, doi.org/10.1016/j.epsl.2016.08.044
- Haqq-Misra, J., Kopparapu, R. K., Batalha, N. E., Harman, C. E., Kasting, J. F. 2016, Limit Cycles Can Reduce the Width of the Habitable Zone, Astrophys. J., doi.org/10.3847/0004-637X/827/2/120
- Batalha, N., Domagal-Goldman, S. D., Ramirez, R., Kasting, J. F. 2015, Testing the early Mars H2-CO2 greenhouse hypothesis with a 1-D photochemical model, Icarus, doi.org/10.1016/j.icarus.2015.06.016

Invited Talks, Seminars, Panels & Colloquia

- **Apr. 2026**: Habitable Worlds Observatory Physical Parameter Workshop
- **Mar. 2026**: Caltech/IPAC Planetary Lunch Seminar
- **Nov. 2024**: Space Telescope Science Institute Colloquium, STScI
- **Nov. 2023**: Center for Interdisciplinary Exploration and Research in Astrophysics Colloquium, Northwestern University
- **Jul. 2023**: Sagan Summer Workshop `PICASO` hands on session and speaker

- **Jun. 2023**: ExoClimes Keynote
- **Apr. 2023**: Department of Astrobiology Colloquium, University of Washington
- **Feb. 2023**: EMAC Workshop: Open Access Exoplanet Modeling & Analysis Tools
- **Dec. 2022**: AGU Fall Meeting: The Future is Open Panel
- **Oct. 2022**: Bay Area Open Science Meeting
- **Aug. 2022**: ASA-HITRAN
- **May 2022**: Exoplanets IV Splinter Session: Enabling Future Comparative Exoplanetology
- **Dec. 2021**: UC Berkeley Center for Integrative Planetary Science Seminar
- **Nov. 2021**: Department of Astrophysics Colloquium, University of California Santa Cruz
- **Oct. 2021**: SACNAS: Exploring the Universe with NASA Astrophysics
- **Aug. 2021**: European Southern Observatory: Atmospheres, Atmospheres! Do I look like I care about atmospheres?
- **Aug. 2021**: NASA Ames Summer Series
- **July 2021**: Sagan Summer Workshop
- **June 2021**: Scialog: Signatures of Life in the Universe
- **Apr. 2021**: College of Science Seminar Series, San Jose State University
- **Nov. 2020**: Astronomy & Astrophysics Colloquium, Caltech Institute of Technology
- **July 2020**: Sagan Summer Workshop
- **Dec. 2019**: OWL @ ETH - paving the way to the atmospheric characterization of terrestrial exoplanets
- **Dec. 2019**: Department of Astronomy Colloquium, University of Michigan
- **Nov. 2019**: Carnegie Observatory Colloquium, Pasadena, CA
- **Jul. 2019**: Moonshots and Earthshots in the Search for Life Beyond Earth, Green Bank, WV
- **Dec. 2018**: Department of Astrobiology Colloquium, University of Washington
- **Nov. 2018**: Department of Space Sciences Planetary Lunch Seminar, Cornell University
- **Nov. 2018**: Stars and Planets Seminar Series, Harvard Center for Astrophysics
- **Oct. 2018**: Department of Astronomy & Astrophysics, University of California Santa Cruz
- **Oct. 2018**: Department of Physics Colloquium, University of California Merced
- **Jun. 2018**: Panelist at Emerging Researchers in Exoplanets Symposium
- **Jun. 2018**: Planetary Exploration Group Seminar, JHU Applied Physics Lab
- **Feb. 2018**: George Mason University Observatory Public Lecture, Fairfax, VA
- **Jul. 2017**: Enabling Transiting Exoplanet Observations with JWST Workshop, Space Telescope Science Institute
- **Feb. 2017**: School of Earth and Space Exploration Seminar, Arizona State University
- **Aug. 2016**: Planetary Systems: A Synergistic View, Quy Nhon, Vietnam
- **Aug. 2016**: Department of Terrestrial Magnetism Colloquium, Carnegie Institute
- **Mar. 2016**: Planetary Lunch Seminar, Goddard Space Flight Center
- **Mar. 2016**: Planetary Lunch Seminar, Center for Exoplanets and Habitable Worlds, The Pennsylvania State University
- **Feb. 2016**: Seminar, Jet Propulsion Laboratory
- **May 2015**: Special Seminar to The Pennsylvania State Board of Visitors
- **May 2015**: Special Seminar to The Pennsylvania State Dean of Eberly College of Science Advisory

Committee

Contributed Talks

- **Sept. 2019**: Bay Area Exoplanet Meeting, NASA Ames, CA
- **Aug. 2019**: Extreme Solar Systems IV, Reykjavik, Iceland
- **Dec. 2018** : Bay Area Exoplanet Meeting, NASA Ames, CA
- **Sept. 2018**: Bay Area Exoplanet Meeting, NASA Ames, CA
- **Jul. 2018**: Exoplanets II, Cambridge, UK
- **May. 2018**: Chesapeake Bay Area Exoplanet Meeting, Carnegie DTM, MD
- **Jan. 2018**: Winter AAS Conference, Washington DC
- **Jan. 2017**: Winter AAS Conference, Grapevine, Texas
- **Oct. 2016**: Division of Planetary Sciences Conference, Pasadena, CA
- **Jan. 2014** : Winter AAS Conference, Washington, DC